

MedPlusMart Capstone Testing Plan

End-to-End QA Automation Testing Plan for MedPlusMart Website

Project Information

Field	Details
Project Type	E-Commerce Healthcare Website QA Automation Testing
Selected Website	MedPlusMart
Website URL	https://www.medplusmart.com
Automation Tool	Playwright
Programming Language	JavaScript
Framework Design	Page Object Model (POM)
Reporting Tools	HTML Reports, Allure Reports
CI/CD Integration	GitHub Actions
Minimum Requirement	8 modules with 15 test cases each
Planned Test Coverage	120+ Functional Test Cases

Project Overview

This capstone project focuses on automating major user workflows of the MedPlusMart healthcare e-commerce application using Playwright with JavaScript. The project is designed from the perspective of a QA Automation Engineer and aims to validate the stability, functionality, usability, and reliability of important customer-facing features.

The automation framework will cover real-world workflows such as user registration, login, medicine search, product browsing, cart functionality, checkout flow, address management, order handling, and account management.

The complete automation framework will be organized module-wise using the Page Object Model design pattern for better maintainability and scalability.

Reason for Selecting MedPlusMart

The MedPlusMart website was selected for this capstone project because:

- It is a real-world healthcare e-commerce application.
 - The website contains multiple customer-facing modules suitable for automation.
 - It supports practical online shopping workflows.
 - The application includes form validations, product management, authentication, cart handling, and checkout features.
 - The platform provides enough functional coverage to create more than 120 automation test cases.
 - The website is suitable for demonstrating Playwright automation skills using JavaScript.
 - The application provides responsive navigation and realistic user journeys.
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Project Objectives

The main objectives of this capstone project are:

- Design and implement a Playwright automation framework using JavaScript.
 - Prepare more than 120 functional test cases.
 - Validate major user workflows in the MedPlusMart application.
 - Automate positive, negative, regression, and validation scenarios.
 - Generate HTML and Allure execution reports.
 - Implement reusable Page Object Model architecture.
 - Integrate automated test execution with GitHub Actions.
 - Improve test maintainability and execution efficiency.
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Scope of Testing

The scope of this project is limited to customer-facing frontend functionalities available on the MedPlusMart website.

Scope Details

In Scope	Out of Scope
User Registration and Login	Backend Database Validation
Product Search and Product Details	Admin Panel Testing
Medicine Categories and Navigation	Performance Testing
Cart and Quantity Management	Security Testing
Address Management	Load Testing
Order Placement Flow	API Performance Benchmarking
Account and Profile Features	Real Payment Transactions
UI Validations	Server Configuration Changes

Testing Coverage

- Backend Database Validation
- Admin Panel Testing
- Performance Testing
- Security Testing
- Load Testing
- API Performance Benchmarking
- Real Payment Transactions
- Server Configuration Changes

Testing Types Covered

Testing Type	Purpose
Functional Testing	Verify application functionality against expected behavior
Validation Testing	Validate mandatory fields and error messages

Testing Type	Purpose
Negative Testing	Ensure invalid data is handled properly
Regression Testing	Confirm existing features work after updates
UI Testing	Verify visibility, layout, labels, and UI elements
Navigation Testing	Validate menus, redirects, and page movement
End-to-End Testing	Validate complete user workflows
Smoke Testing	Verify core functionalities before execution

Services and Modules Covered

The automation testing framework will cover the following major services of the MedPlusMart application. Each service includes functional validations, UI validations, navigation checks, negative scenarios, and regression coverage.

Service / Module	Module Description	Severity	Planned Test Cases
User Authentication	Covers registration, login, logout, forgot password, and user session validation workflows.	High	15
Product Search and Catalog	Covers medicine search, categories, filters, product details, and product navigation.	High	15
Cart Management	Covers add to cart, quantity update, remove item, cart persistence, and cart calculations.	High	15
Checkout and Payments	Covers checkout flow, delivery details, order summary, and order placement workflows.	Critical	15
Address Management	Covers adding, editing, deleting, and validating delivery addresses.	Medium	15
Account Management	Covers profile updates, password changes, saved addresses, and order history.	Medium	15

Service / Module	Module Description	Severity	Planned Test Cases
Navigation and UI Testing	Covers menus, redirects, responsiveness, labels, and UI visibility validations.	Medium	15
Customer Support Features	Covers help pages, contact forms, notifications, and support-related features.	Low	15

Total Planned Test Cases: 120+

Detailed Module Coverage

User Authentication

The authentication module validates user access and account security workflows.

Product Search and Catalog

This module validates product browsing and search functionality.

Cart Management

The cart module validates shopping cart operations.

Checkout and Payments

This module validates end-to-end order placement workflows.

Address Management

The address module validates address handling features.

Account Management

This module validates account-related customer functionalities.

Navigation and UI Testing

This module validates application navigation and user interface behavior.

Customer Support Features

This module validates customer assistance functionalities.

Planned Project Structure

MedPlusMart_Automation_Project/

```
|
├── tests/
|   ├── authentication.spec.js
|   ├── productCatalog.spec.js
|   ├── cart.spec.js
|   ├── checkout.spec.js
|   ├── address.spec.js
|   ├── account.spec.js
|   ├── navigation.spec.js
|   └── support.spec.js
|
├── pages/
|   ├── LoginPage.js
|   ├── ProductPage.js
|   ├── CartPage.js
|   ├── CheckoutPage.js
|   └── AccountPage.js
|
├── test-data/
├── reports/
├── playwright.config.js
└── package.json
```

Test Data Strategy

Dummy data will be used throughout the project to avoid the usage of real customer information.

Planned Test Data

- Dummy usernames
- Dummy email addresses
- Dummy mobile numbers
- Dummy delivery addresses
- Sample product names
- Test passwords

Data Handling Strategy

- Dynamic email generation
 - Reusable JSON test data
 - Separate data files for maintainability
 - Configurable environment setup
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Planned Capstone Workflow

- Finalize testing scope and services
 - Prepare module-wise test cases
 - Design the automation framework
 - Implement Playwright project structure
 - Develop automation scripts service-wise
 - Execute and debug failed scenarios
 - Generate execution reports
 - Prepare screenshots and execution evidence
 - Organize final submission documents
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Final Deliverables

Deliverable	Description
Testing Plan Document	Complete project planning document
Test Case Document	Module-wise test cases
Automation Framework	Playwright automation scripts
Test Data Files	Reusable dummy data
HTML Reports	Execution reports
Allure Reports	Detailed reporting dashboard
Evidence Folder	Screenshots and trace files
Final Summary	Project execution summary

Conclusion

The MedPlusMart automation testing project is suitable for a QA automation capstone because it provides multiple real-world healthcare e-commerce workflows that can be automated effectively using Playwright with JavaScript.

The project demonstrates:

- Automation framework development
- Functional testing knowledge
- End-to-End testing skills
- Reporting implementation
- Page Object Model architecture
- CI/CD integration
- Real-time workflow validation

This capstone project helps showcase practical QA Automation Engineering skills using industry-standard testing practices and modern automation tools.